

Lecture 13

Good Data, Bad Outcomes

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"After analyzing all your data, I think we can safely say that none of it is useful."

Last Time

- ❖ Potential issues with data
 - Missing values
 - Formatting inconsistencies
 - Duplication
 - Outliers
- ❖ Importance of documenting data decisions

Today

Ways analyses can go wrong even with clean data

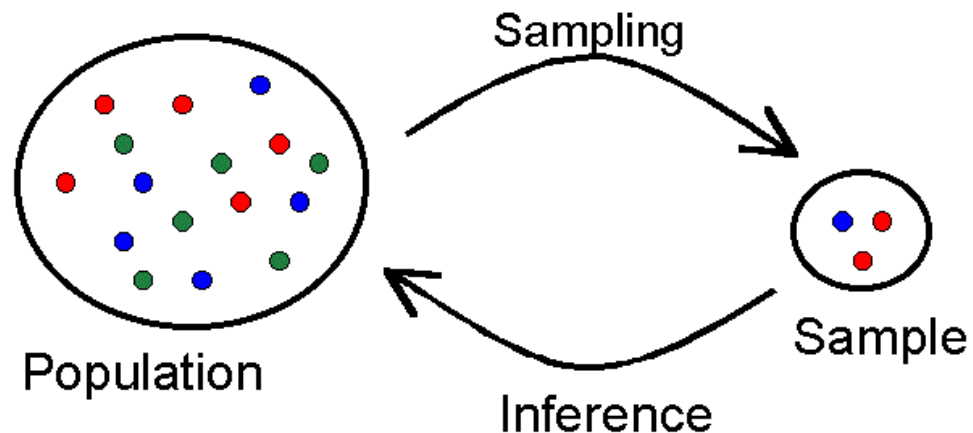
- ❖ Survivorship Bias
- ❖ Selection bias
- ❖ Goodhart's Law
- ❖ Confounding
- ❖ Simpson's Paradox

Inference

- ❖ Drawing conclusions about the world from our dataset

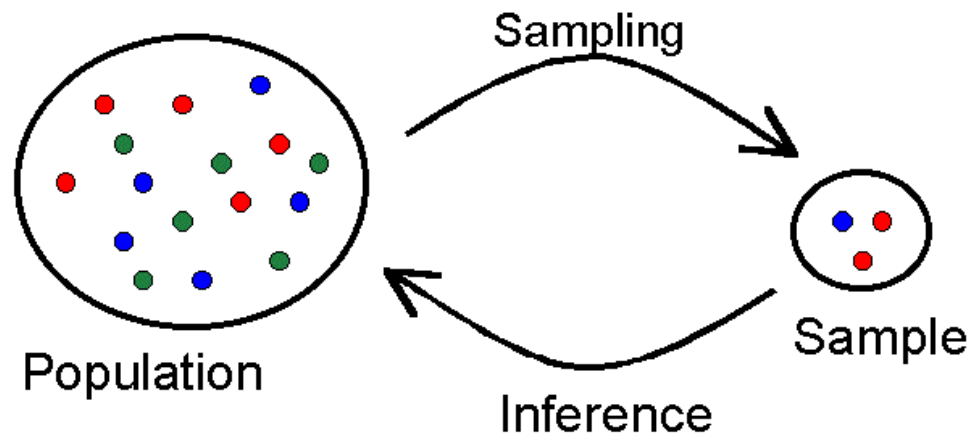
Examples:

- ❖ Given the heights of some Allegheny students, we make judgments about the heights of all students
- ❖ Use past customer behavior to predict future customer behavior



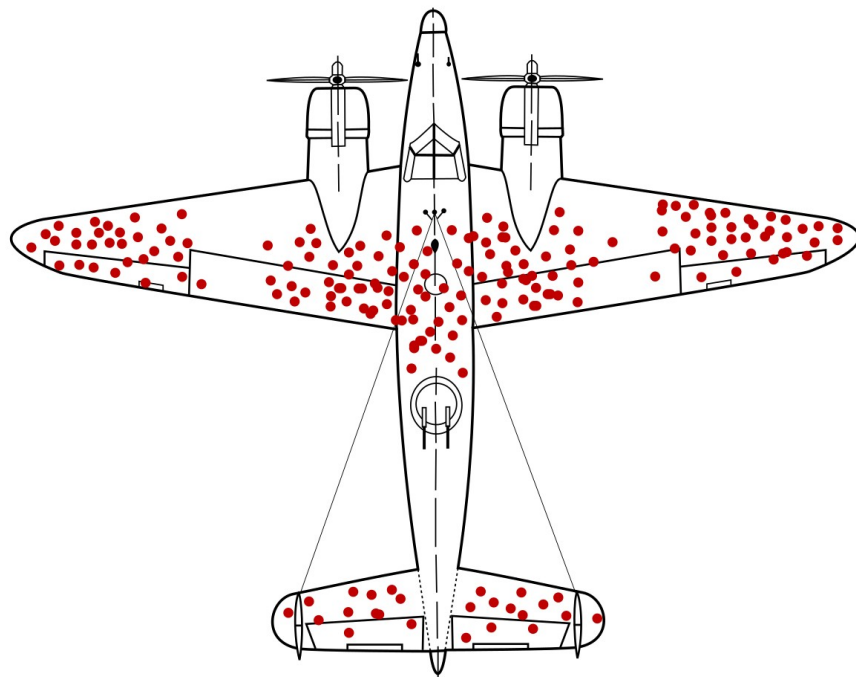
Inference

- ❖ Drawing conclusions about the world from our dataset
- ❖ An inference is valid if what's going on in the sample corresponds to what's going on the population
- ❖ Validity is determined by
 - Methods used to reason about data
 - Provenance of our data



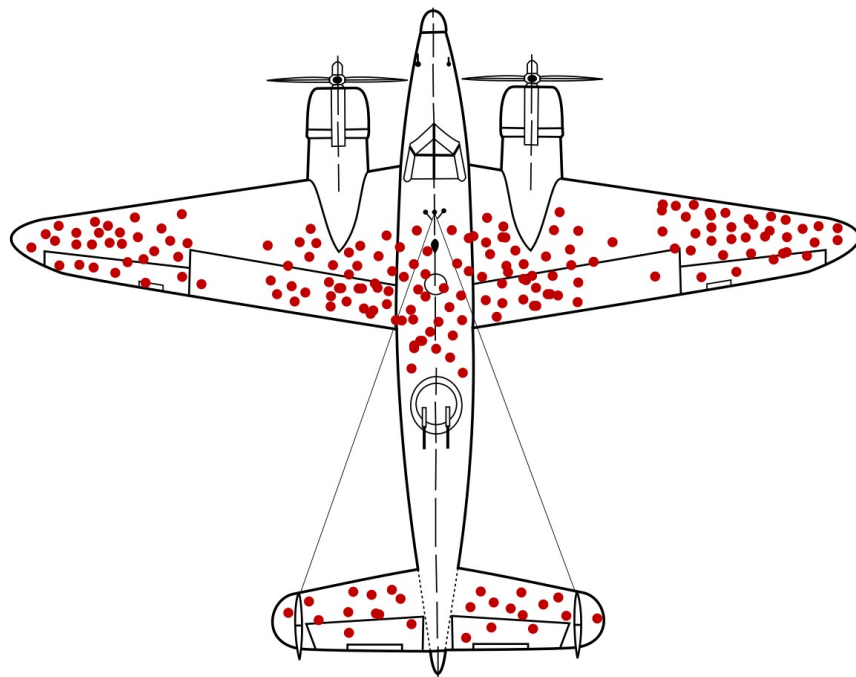
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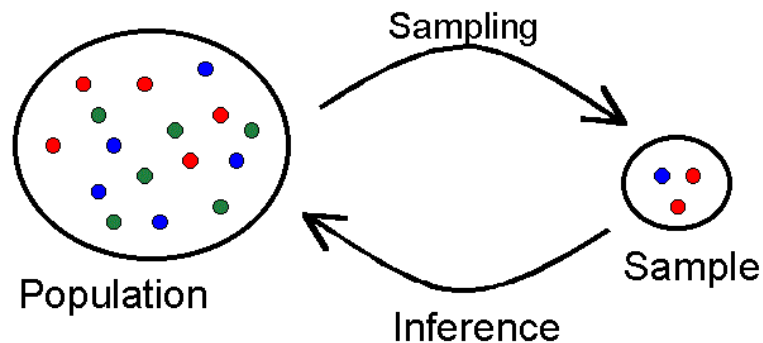
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- ❖ Planes damaged in other areas did not make it back!



Selection Bias

- ❖ Data collection is rarely random
- ❖ This may cause a mismatch between our data and the **target population**
 - Target population is the population you want to reason about
 - This mismatch is called **bias**
- ❖ Bias can lead to false conclusions about the target population



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Target Pop: all online customers

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Is there bias?

- ❖ Potentially!
- ❖ What about people who didn't or couldn't buy online because the website was too difficult to use?

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 - Target variable: Is the patient healthy?
 - Proxy measure: Body mass index (BMI)
- ❖ Why can this be an issue?
 - Patient health may correlate with BMI, but there are unhealthy low-BMI patients
 - There's a mismatch between the proxy and the target

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Known as Goodhart's Law:

- ❖ "When a measure becomes a target, it ceases to be a good measure."

Next Time

- ❖ Finishing up biases and other issues